

Study Questions: Planning

1. What is a multi-step policy and how does it differ from the single-step policies used in earlier modules?
2. Write the formula for total EFE of a multi-step policy $\pi = [a_0, a_1, \dots, a_{T-1}]$.
3. How does the agent predict future state distributions? Write the recursive formula using the B-matrix.
4. What is Marginal Message Passing (MMP) and why is it needed for deep temporal models?
5. In `run_mmp()`, what does the `beliefs` return value contain? How many belief vectors are returned for a 3-step policy?
6. Explain the computational tradeoff of increasing temporal depth T . How does the number of possible policies scale with T ?
7. For a 4-action system with $T = 3$, how many exhaustive policies exist? How could you reduce this number?
8. Write code to define multi-step policies for a T-maze agent that can plan 2 steps ahead (first go to cue, then go to arm).
9. Why does the T-maze require $T \geq 2$ for the agent to reliably find the reward? What fails with $T = 1$?
10. How would you encode a 3×3 gridworld as a `DiscreteEnvironment`? What shape would the B-tensor have?
11. How are obstacles (walls) represented in the B-matrix of a gridworld? What transition does a wall create?
12. What does `plot_gridworld()` visualize? What arguments does it accept?

13. Write the full code for evaluating a multi-step policy's total EFE by unrolling predicted states across all timesteps.
14. How does MMP differ from simple forward inference? What additional information does backward message passing provide?
15. In a gridworld with a distant reward (10 steps away), what minimum temporal depth T is needed? What practical problem arises?
16. How does `plot_simulation_dashboard()` organize its 5 panels? What is shown in each panel?
17. Explain how hierarchical planning could reduce the policy space. What are "abstract policies" in this context?
18. Compare a $T = 1$ agent and a $T = 5$ agent on the same gridworld. What differences in behavior would you expect?
19. What is the relationship between planning depth T and the exploration–exploitation tradeoff? Does deeper planning favor exploitation?
20. Design a delayed-reward task where a $T = 1$ agent fails but a $T = 3$ agent succeeds. Specify the states, observations, actions, C-vector, and the necessary policy structures.